

$$\lambda^2 - 6\lambda + 13 = 0$$

$$\lambda = 3 \pm 2i$$

$$\begin{aligned} y_p &= a e^{-2t} \\ y_p' &= -2a e^{-2t} \\ y_p'' &= 4a e^{-2t} \end{aligned}$$

$$y_p'' - 6y_p' + 13y_p = e^{-2t}$$

$$\Rightarrow (4a + 12a + 13a)e^{-2t} = e^{-2t}$$

$$\Rightarrow a = 1/29$$

Name

$$y = y_c + y_p$$

$$y' = 3c_1 e^{3t} \cos(2t) + 2c_1 e^{3t} \sin(2t)(-1) + 3c_2 e^{3t} \sin(2t) + 2c_2 e^{3t} \cos(2t) - 2/29 e^{-2t}$$

■ For the ODE $y'' - 6y' + 13y = e^{-2t}$

■ Write down the complementary sol.

$$y_c(t) = c_1 e^{3t} \cos(2t) + c_2 e^{3t} \sin(2t)$$

$$y(0) = 2 \Rightarrow c_1 e^0 \cos(0) + c_2 e^0 \sin(0) = 2$$

$$c_1 + 0c_2 = 2$$

■ Compute a particular sol.

$$y_p(t) = 1/29 e^{-2t}$$

$$y'(0) = 3 \Rightarrow 3c_1 e^0 \cos(0) - 2c_1 e^0 \sin(0) + 3c_2 e^0 \sin(0) + 2c_2 e^0 \cos(0) = 3$$

$$3c_1 + 2c_2 = 3$$

$$c_1 + 0c_2 = 2 - 1/29$$

$$3c_1 + 2c_2 = 3 + 2/29$$

■ The Aug Mat for $y(0) = 2, y'(0) = 3$ is

$$\begin{bmatrix} 1 & 0 & 1 & 2 - 1/29 \\ 3 & 2 & 1 & 3 + 2/29 \end{bmatrix}$$

1. For the ODE $y'' - 5y' - 14y = \sin(t)$

1.1. Write down the complementary sol.

$$y_c(t) =$$

1.2. Compute a particular sol.

$$y_p(t) =$$

1.3. The Aug Mat for $y(0) = 1, y'(0) = 4$ is

$$[\quad]$$

$$\begin{cases} \lambda^2 - 4\lambda + 9 = 0 \\ \lambda = 2 \pm \sqrt{5}i \end{cases}$$

$$y_p = a \cos(2t) + b \sin(2t)$$

$$y_p' = -2a \sin(2t) + 2b \cos(2t)$$

$$y_p'' = -4a \cos(2t) - 4b \sin(2t)$$

$$y_p'' - 4y_p' + 9y_p = (5a - 8b) \cos(2t) + (8a + 5b) \sin(2t) = 0 \cos(2t) + 3 \sin(2t)$$

■ For the ODE $y'' - 4y' + 9y = 3 \sin(2t)$ with

■ Write down the complementary sol.

$$y_c(t) = c_1 e^{2t} \cos(\sqrt{5}t) + c_2 e^{2t} \sin(\sqrt{5}t)$$

■ Compute a particular sol.

$$y_p(t) = \frac{24}{89} \cos(2t) + \frac{15}{89} \sin(2t)$$

■ The Aug Mat for $y(0) = 2, y'(0) = -3$ is

$$\left[\begin{array}{cc|c} 1 & 0 & 2 - \frac{24}{89} \\ 2 & \sqrt{5} & -3 - 2\left(\frac{15}{89}\right) \end{array} \right]$$

$$\left[\begin{array}{cc|c} 5 & -8 & 0 \\ 8 & 5 & 3 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{24}{89} \\ 0 & 1 & \frac{15}{89} \end{array} \right]$$

$$y = y_c + y_p$$

$$y(0) = c_1 + 0c_2 + \frac{24}{89} = 2$$

$$c_1 + 0c_2 = 2 - \frac{24}{89}$$

$$y' = 2c_1 e^{2t} \cos(\sqrt{5}t) + c_1 \sqrt{5} \sin(\sqrt{5}t) + 2c_2 e^{2t} \sin(\sqrt{5}t) + c_2 \sqrt{5} \cos(\sqrt{5}t) + 2\left(\frac{24}{89}\right) \sin(2t) + 2\left(\frac{15}{89}\right) \cos(2t)$$

$$y'(0) = -3 = 2c_1 + \sqrt{5}c_2 + 2\left(\frac{15}{89}\right)$$

$$2c_1 + \sqrt{5}c_2 = -3 - 2\left(\frac{15}{89}\right)$$

2. For the ODE $y'' - 2y' + 5y = 2 \cos(3t)$

2.1. Write down the complementary sol.

$$y_c(t) =$$

2.2. Compute a particular sol.

$$y_p(t) =$$

2.3. The Aug Mat for $y(0) = -2, y'(0) = 7$ is

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